

week 3: ROC curves, Naive Bayes and Entropy

<http://mlvu.github.io>

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1 Exam questions

At this point in the course, it's a good idea to have a look at the practice exam. The homework prepares you for the exam, but the format of the homework questions is different from that of the exam questions. Here are some questions you should be able to answer based on the material covered so far.

Canvas offers a complete practice exam, a documentation of how the exam works and two Canvas quizzes which randomly generate exams.

1. What is the advantage of gradient descent over random search?
 - A In gradient descent, parallel searches are allowed to communicate.
 - B Gradient descent is less likely to get stuck in local minima.
 - C Gradient descent computes the direction of steepest descent, random search approximates it.
 - D Gradient descent is easier to parallelise.
2. Why is accuracy a bad loss function to use in gradient descent?
 - A It is expensive to compute.
 - B It makes the gradient zero almost everywhere.
 - C It is unreliable in situations with high class imbalance.
 - D The confidence interval is high on small test sets.
3. I'm performing spam classification. I represent each email by three numbers: how often the word *pill* occurs, how often the word *hello* occurs and how often the word *congratulations* occurs. What are these three attributes called?
 - A The instances
 - B The classes
 - C The features
 - D The principal components

Here we see the derivation of the gradient of the squared-error loss for linear regression.

$$\frac{\partial \frac{1}{2} \sum_i (f(x_i) - y_i)^2}{\partial b} = \frac{1}{2} \frac{\partial \sum_i (x_i w + b - y_i)^2}{\partial b} \quad (1)$$

$$= \frac{1}{2} \sum_i \frac{\partial (x_i w + b - y_i)^2}{\partial b} \quad (2)$$

$$= \frac{1}{2} \sum_i \frac{\partial (x_i w + b - y_i)^2}{\partial (x_i w + b - y_i)} \frac{\partial (x_i w + b - y_i)}{\partial b} \quad (3)$$

$$= \sum_i (x_i w + b - y_i) \frac{\partial (x_i w + b - y_i)}{\partial b} \quad (4)$$

$$= \dots \quad (5)$$

4. To get from line (1) to line (2), we use the
A Chain rule **B** Product rule **C** Exponent rule **D** Sum rule
5. To get from line (2) to line (3), we use the
A Chain rule **B** Product rule **C** Exponent rule **D** Sum rule
6. In line 5, the correct result is
A $\sum_i (x_i^2 w + b - y_i)$
B $\sum_i x_i (x_i w + b - y_i)$
C $\sum_i (x_i w + b - y_i)$
D $\sum_i (x_i w + b - y_i)^2$

2 ROC Curves and confusion matrices

In this section, we'll look at evaluation metrics. This section covers the lecture *Evaluation* and section 2.2 of Peter Flach's *Machine Learning* (see the PDF on Canvas).

As a running example, we will use the following classification dataset:

	x_1	x_2	label
a	1	0	Neg
b	2	0	Pos
c	3	0	Pos
d	$1/2$	1	Pos
e	2	2	Pos
f	0	3	Neg
g	1	3	Neg
h	2	3	Neg

If you have trouble with the exercises below, we recommend drawing the data in its feature space, and drawing the decision boundary of each classifier.

2.1 A linear classifier

question 1: As a first classifier, c_{lin} , we will use a diagonal line crossing all the points where $x_1 = x_2$. Points above this line will be classified as Neg, points below or on the line as Pos. How do we describe this classifier mathematically? Fill in the blanks:

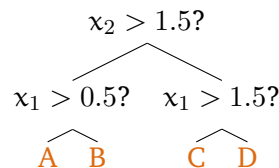
$$c_{\text{lin}}(x_1, x_2) = \begin{cases} \text{Pos} & \text{if } \dots \\ \text{Neg} & \text{otherwise} \end{cases} .$$

c_{lin} is a simple classifier. Let's try a more complex one before we move on.

question 2: Let c_2 be the classifier whose decision boundary crosses the points $(0, 1)$ and $(1, 1.5)$, with the class Neg above the line. Draw this line first. How do we define this classifier?

$$c_2(x_1, x_2) = \begin{cases} \text{Pos} & \text{if } \dots \\ \text{Neg} & \text{otherwise} \end{cases}$$

question 3: We will compare the linear classifier c_{lin} with a *decision tree* classifier, c_{tree} . Here is the decision tree:



If the inequality on the node is true, move right, otherwise move left.

This tree divides the feature space into four **segments** (A, B, C and D). To each segment we assign the majority class in that segment (using Pos for a draw). Label the leaves A, B, C and D with classes.

question 4: In this exercise we don't do any training (we just evaluate the given classifiers), so we'll evaluate all metrics on a single given dataset.

If you do train, you would *split* your dataset. You would compute your evaluation metrics on the ... during hyperparameter selection and on the ... during final testing.

Evaluating on the training data is the worst sin you can commit in Machine Learning. Can you think of a situation when you would still want to *compute* the training loss?

question 5: Give the *confusion matrices* for classifiers c_{lin} and c_{tree} .

question 6: From the confusion matrices, we can compute several metrics. Compute the accuracy, precision, recall, true positive rate and false positive rate

To plot a ROC and precision/recall curves, and to compute AUC metrics, we need to turn our simple binary classifier into a *ranking* classifier. That is, we don't just want classifications, we want it to rank the data from most Neg to most Pos. How we do this depends on the classifier.

question 7: How do we turn a linear classifier into a ranking classifier? Give the ranking that c_{lin} assigns to the training data.

question 8: How do we turn a decision tree classifier into a ranking classifier? Give the ranking that c_{tree} assigns to the training data.

We can't tell whether our ranking is perfect, because the training data doesn't give us a correct *ranking*, only a correct *labeling*. But for some pairs of instances we *do* know how they should be ranked: all pairs of positive and negative examples.

We can visualize this in a matrix with **negative** instances on the horizontal axis, and **positive** instances on the vertical, both sorted with the most positive (according to the classifier's ranking) in the bottom left corner. The

matrix is usually colored **green** for pairs that are ranked correctly, **red** for pairs ranked incorrectly, and **yellow** for pairs that are ranked equal.

question 9:

1. Draw the coverage matrices of c_{lin} and c_{tree} .
2. How many ranking errors does each classifier make on the training data?
3. If we have class imbalance (one class has more instances than the other), how can we tell by the coverage matrix?

question 10: How do we convert a coverage matrix to an **ROC plot**? How does the green area in a coverage matrix relate to the ROC plot? How do they differ?

3 preliminaries: probability

If the exercises below give you any trouble we recommend that you follow the links provided to brush up a little.

question 11:

1. In a few sentences, explain the difference between the Frequentist and the Bayesian interpretation of probability.
2. What is the difference between a sample space and an event space?
3. Jack and Jill have a dataset of biometric data of Dutch men, and they fit a normal distribution to their height. They get a mean of 183.8 cm. Jack suggest that this means that the probability of seeing a Dutch man this tall is higher than any other height. Jill disagrees: she suggests that the more precisely we state the mean (with more digits after the decimal point) the smaller the probability will get. For the exact mean, the probability of seeing someone with that exact height is zero. Jack points to the shape of the normal distribution: it peaks at the mean, surely that means that that is where the most probable outcome is. Who is right?

question 12: In the following, p is a probability function and A and B are random variables. Which of the following are true?

1. Joint probability is symmetric: $p(A, B) = p(B, A)$.
2. Conditional probability is symmetric $p(A | B) = p(B | A)$.
3. Two random variables X and Y are conditionally independent on a third Z . Once we know X and Y , we also know the value of Z .
4. Two random variables X and Y are conditionally independent on a third Z . Once we know Z , also knowing X will tell us nothing extra about Y .

question 13: Assume that the probability that a given patient has diabetes is 0.01. We have a test for diabetes with a false positive rate of 0.05: if a patient has no diabetes, the test diagnoses it 5% of the time. The false negative rate is 0.1. You are a doctor, and you administer the test to a patient (knowing nothing else). The test says she has diabetes. What is the probability that she doesn't? Hint: this is a question about Bayes' rule. Reflect on the result. Is this what you would've expected? If not, where does the unexpected result come from?

More practice? Follow these links:

1. <https://seeing-theory.brown.edu/> (especially this one)
2. <https://betterexplained.com/articles/a-brief-introduction-to-probability-statistics/>
3. <https://www.khanacademy.org/math/probability/probability-geometry/probability-basics/a/probability-the-basics>
4. <http://dept.stat.lsa.umich.edu/~moulib/probrefresh.pdf>

4 Naive Bayes

The following dataset represents a spam classification problem: we observe 8 emails and measure two binary features with the values **True** and **False**. The first is **True** if the word “pill” occurs in the e-mail and **False** if it doesn't, and the second is **True** if the word “meeting” occurs and **False** if it doesn't.

“pill”	“meeting”	label
T	F	Spam
T	F	Spam
F	T	Spam
T	F	Spam
F	F	Ham
F	F	Ham
F	T	Ham
T	T	Ham

question 14: We will build a naive Bayes classifier for this data. What is the defining property of naive Bayes? Why is it called “naive”?

question 15: We build a naive Bayes classifier on this data, as described in the lecture. We get an email that contains both words. Which class does the classifier assign?

question 16: Which probabilities does the classifier assign to each class?
To improve our accuracy, we add another feature:

“pill”	“meeting”	“hello”	label
T	F	F	Spam
T	F	F	Spam
F	T	F	Spam
T	F	F	Spam
F	F	F	Ham
F	F	F	Ham
F	T	F	Ham
T	T	T	Ham

question 17: For the class Spam, there are no emails recorded that contain the word “hello”. Why is this a problem?

What solution is suggested in the slides?

question 18: Implement this solution, and give the class probabilities for an email containing all three words.

5 Entropy

We define two probability distributions p and q on a set of four outcomes $\{a, b, c, d\}$.

$$\begin{array}{cccc} p(a) & p(b) & p(c) & p(d) \\ \hline 1/4 & 1/4 & 1/4 & 1/4 \end{array}$$

$$\begin{array}{cccc} q(a) & q(b) & q(c) & q(d) \\ \hline 1/2 & 1/4 & 1/8 & 1/8 \end{array}$$

question 19: Could you simulate sampling from these distributions using coinflips as described in the lecture?

question 20: Compute the entropy of p and q .

The entropy of q is lower than the entropy of p . What does this tell you about the difference between the two distributions?

question 21: In the definition of entropy that we use (information entropy), the logarithms have base 2 (i.e. $\log_2$ instead of $\log_{10}$ or $\ln$). This follows directly from our decision to model probability distributions with coinflips. How?